

number		
4(a)(i)	The only correct answer is B <i>A is incorrect because P is the vena cava</i> <i>C is incorrect because R is the aorta and P is the vena cava</i> <i>D is incorrect because R is the aorta</i>	(1)

number		
4(a)(ii)	The only correct answer is A <i>B is incorrect because R is the aorta</i> <i>C is incorrect because Q is the pulmonary artery</i> <i>D is incorrect because Q is the pulmonary artery and R is the aorta</i>	(1)

Question number	Answer	Mark
4(a)(iii)	The only correct answer is C <i>A is incorrect because P is the vena cava and the coronary artery branches from the aorta</i> <i>B is incorrect because Q is the pulmonary artery and the coronary artery branches from the aorta</i> <i>D is incorrect because S is the pulmonary vein and the coronary artery branches from the aorta</i>	(1)

number			
4(b)(i)	Any two from: surface area (of diffusion surface) concentration {gradient / difference} membrane permeability temperature pressure type of molecule	ACCEPT velocity of blood flow concentration {gradient / difference} of appropriate named molecule ACCEPT size, charge	(1)

Question number	Answer	Additional guidance	Mark
4(b)(ii)	Fick's (law) (1)		(1)

number																							
4(c)(i)	An answer that makes reference to the following points: <table border="1"> <thead> <tr> <th>Diffusion distance</th><th>Diffusion distance / μm</th><th>Time taken to diffuse the distance</th><th>Time taken to diffuse the distance / mseconds</th></tr> </thead> <tbody> <tr> <td></td><td></td><td></td><td></td></tr> <tr> <td></td><td></td><td></td><td></td></tr> <tr> <td></td><td></td><td></td><td> 5 000 / 5 000.0 / 5×10^3 </td></tr> <tr> <td></td><td> 1000 / 1000.0 / 1×10^3 </td><td></td><td> 498 000 / 498 000.0 / 4.98×10^5 / 500 000 / 5×10^5 </td></tr> </tbody> </table>	Diffusion distance	Diffusion distance / μm	Time taken to diffuse the distance	Time taken to diffuse the distance / mseconds												5 000 / 5 000.0 / 5×10^3		1000 / 1000.0 / 1×10^3		498 000 / 498 000.0 / 4.98×10^5 / 500 000 / 5×10^5	1 mark = 1 or 2 correct values 2 marks = all correct	(2)
Diffusion distance	Diffusion distance / μm	Time taken to diffuse the distance	Time taken to diffuse the distance / mseconds																				
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	1000 / 1000.0 / 1×10^3		498 000 / 498 000.0 / 4.98×10^5 / 500 000 / 5×10^5																				

Question	Answer	Additional guidance	Mark
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number			
4(c)(ii)	An explanation that makes reference to three of the following points: - to supply the {cells / tissues} with {oxygen / glucose} (for respiration) (1) - to remove the carbon dioxide (produced) (1) - a small (diffusion) distance results in a faster the rate of {diffusion / exchange} (1) - credit use of data (with at least one comparable unit) to support mp 3 (1)	ACCEPT named cells / tissues e.g. muscle ACCEPT correct reference to time for diffusion for rate converse IGNORE efficiently e.g. it takes 0.5 milliseconds for 1 um but 50 milliseconds for 10um more than {10 / 100} μm away the rate of diffusion would not be fast enough	(3)